Introduction. Number systems. Codes. Errors

1. Number systems:

Def. A number system is a given set of rules for representing and encoding information using a set of symbols called digits.

Classification:

1. Non-positional: The value/weight of a digit does not depend on its position within the number (Roman n.).
2. Positional: Each digit has an associated weight based on a base b.

Obs. A number N expressed in base b is written:

$$(N)_b = c_{n-1} \cdot b^{n-1} + c_{n-2} \cdot b^{n-2} + \ldots + c_0 b^0 + c_{-1} b^{-1} + \ldots + c_{-p} b^{-p}$$

b - base of system ($b > 1$)
c_i - digits, $0 \leq c_i \leq b-1$
n - number of digits in the integer part
p - number of digits in the fractional part

Obs. The radix point separates the integer part from the fractional part.

Def. In a given base b, the complement of a digit c is defined as:

$$\bar{c} = (b-1) - c$$

Common number systems:

1. Binary system ($b=2$)

ex. $(1011.101)_2 = 1\cdot 2^3 + 0\cdot 2^2 + 1\cdot 2^1 + 1\cdot 2^0 + 1\cdot 2^{-1} + 0\cdot 2^{-2} + 1\cdot 2^{-3} = 11.62$

2. Octal system ($b=8$)
   - Each octal digit can be encoded using 3 bits

3. Hexa decimal ($b=16$)
   - |——————————//——————————| 4 bits

Conversion between bases:

Case A: $b_1 < b_2$ (smaller to bigger base)

→ the number $(N)_{b_1}$ is expressed as a polynomial using powers of b_1 and evaluated using b_2 arithmetic.

Case B: $b_2 > b_1$ (bigger to smaller base)

→ Calculations are made using b_1 arithmetic:
- Integer part: Repeatedly divide by base b_2. Collect the remainders from right to left until the quotient becomes 0
- Fractional part: Repeatedly multiply by b_2. Collecting integer parts from left to right until the desired accuracy is reached.

ex. $b_1 = 10$ $N = 365.6875$

$b_2 = 2$

$365 \mid 2$

$\vdots \quad \vdots$

$\vdots \quad \vdots$

$\vdots \quad \vdots$

$0 \mid 6875$

$1 \mid 3750$

$0 \mid 7500$

$1 \mid 5000$

$1 \mid 0000$

$\Rightarrow$ Fractional $(N) = .1011$

2. Binary codes for decimal numbers:

Def. Weighted codes are ones in which every bit b_i has an associated weight w_i. The decimal digit value is expressed as:

$$N = b_3 w_3 + b_2 w_2 + b_1 w_1 + b_0 w_0$$

ex. BCD 8421, Aiken 2421, 642-3 code

Obs. 1) A num is self-complementary if the sum of the weights is 9.

2) For any decimal digit A, the code word for its complement $\bar{A} = 9 - A$ is obtained by inverting all bits.

Non-weighted codes:

1. Excess 3 code:

- Obtained by adding the binary value 3 to each BCD 421 code word.

- It is a non-weighted self-complementary code.

2. Gray code
   - A reflected and cyclic code
   - Any two consecutive code words differ by one single bit.
   - An n-bit gray code is constructed by reflecting the (n-1) bit code and appending a leading bit

3. Error detection

Parity bit method:

Consists of appending an extra bit to the leftmost position of the code word:

- Even parity:
  - The parity bit is set to 0 if the nr. of 1s in the code is even
  - The parity bit is set to 1 if the nr of 1s is odd.
- Odd parity:
  - The parity bit is set to 0 if the nr. of 1s is odd
  - The parity bit is set to 1 if the nr. of 1s is even.

Rule: For succesful error detection, both the source and destination must use the same parity type.

Exercises:

1. $(1001101)_4 \rightarrow$
$01000001010001 = (100000.1010001)_2$
$(001)(000)(001)(010)(001) = (10121)_8$
$(0001)(0000)(0101)(0001) = (1051)_{16}$

2. $X = (CABA5)_{16}$

$C = 0000$
$A = 1010$
$B = 1011$
$4 = 0100$
$5 = 0101$

$\Rightarrow X = (0000.1010.1011.0100.0101)_2$

$X = (0022231111)_4$

$X = (01255051)_8$

3.

0	00	000
1	01	001
	11	011
	10	010
		110
		111
		101
		100

11 -> 110
12 -> 111
13 -> 101
14 -> 100
15 -> 0000

4. $273.28125 = ?$

$\Rightarrow$ 1(273.28125) = 100010001

	28125
0	
	56250
1	12500
0	25000
0	50000
1	0

273	2
136	1
68	0
34	0
17	0
8	1
4	0
2	0
1	0
0	1

$\Rightarrow 273.28125 = (100010001.01001)_2$

$= (0100010001.010010)_2 = (10101.102)_4$

$= (421.22)_8 = (111.48)_{16}$